Two explicit Higman–Sims incidence constructions

This component verifies the two constructions in Section 4 of Leemans [1]. It uses the literal Higman–Sims group and the accepted finite coset catalogue, and provides explicit maps between that catalogue and the set constructions. The historical Higman model and its identification with HS are attributed in [1] to Higman and Sims. We do not replace that attribution by a claim of a new construction.

Our retained group input is a faithful permutation model $G \cong \mathrm{HS}$ of order 44 352 000, together with its complete automorphism action of order 88 704 000. The two previously recognized coset families have parabolic orders

$$(2880, 252000, 40320), \qquad (2880, 252000, 252000). \tag{1}$$

Their catalogue identifiers are 35 and 36. These identifiers refer to the acquired catalogue, not to numbering in the historical supplement. We retain the classical identification of the unique Moore graph of valency seven on fifty vertices as the Hoffman–Singleton graph, and its automorphism group $U_3(5) : 2$, as used in [1, Section 4]. The exact finite permutation and graph algorithms used in the certificate are stated with their software inputs; group orders alone are never used to identify an unnamed abstract group.

Theorem 1. *There are sets $\mathcal{P}, \mathcal{C}, \mathcal{Q}$, of respective sizes $176, 1100, 176$, in the retained literal model, with conics $C \in \mathcal{C}$ eight-subsets and quadrics $Q \in \mathcal{Q}$ fifty-subsets of $\mathcal{P}$, satisfying all six Higman properties printed in [1, Section 4.1]. Put $\mathcal{L} = \binom{\mathcal{P}}{2}$.*

1. *The geometry Γ_C on $\mathcal{P}, \mathcal{L}, \mathcal{C}$, with incidence given by containment, is isomorphic to family 35. Its point residue is the Hoffman–Singleton graph, its conic residue is the vertex–edge geometry of K_8, and its line residue is a digon. It has $\mathrm{Aut}(\Gamma_C) = \mathrm{Cor}(\Gamma_C) = G$.*

2. *In Γ_Q on $\mathcal{P}, \mathcal{L}, \mathcal{Q}$, point incidence is containment. A line is incident with a quadric precisely when the quadric contains both conics through its two points. This geometry is isomorphic to family 36. Both point and quadric residues are Hoffman–Singleton graphs, and its line residue is a digon. Its type-preserving group is G, and $\mathrm{Cor}(\Gamma_Q) \cong \mathrm{Aut}(\mathrm{HS})$.*

Both geometries have 61 600 chambers and Borel subgroup S_6. Their line-residue kernels are S_6, and their line stabilizers are split extensions $S_6 : 2^2$. Their point stabilizers, and the quadric stabilizers of Γ_Q, act faithfully as the full $U_3(5) : 2$ on their Hoffman–Singleton residues. The conic stabilizer of Γ_C acts faithfully as S_8 on its eight points.

Literal Higman data and the sixth property

Take the right-coset action of G on a specified subgroup of order 252000. The saved generator images give a homomorphism with trivial kernel on the resulting 176 labelled points. A saved element of the complete ambient automorphism action conjugates the second family's point parabolic to the first family's point parabolic. This actual element, all three transported subgroups, and all point-generator images are checked. Thus the two coordinate systems agree by an explicit map.

The conics and quadrics are the full orbits of the specified base eight-set and fifty-set. Every generator preserves the complete lists and their incidence relations. Each of the four sets of objects (points, lines, conics and quadrics) is a single orbit. Literal subset calculations give fifty quadrics through each point and eight quadrics through each conic. For each of all 15400 point pairs there are exactly two conics C_1, C_2, and

$$\{Q : p, p' \in Q\} = \{Q : C_1 \subseteq Q\} \cup \{Q : C_2 \subseteq Q\}, \qquad |\{Q : C_1 \cup C_2 \subseteq Q\}| = 2. \tag{2}$$

The two quadrics in the last set give a bijection from point pairs to quadric pairs. Conversely, every quadric pair contains exactly two conics; its point intersection is their union, and their intersection has size two. These checks establish properties (i)–(v), including both universal alternatives in (iv) and (v).

For property (vi), fix the retained conic S. For each of its eight points p and each of its eight incident quadrics Q, test both conditions in the source: every conic meeting S in two points including p is contained in Q, and every conic contained in two quadrics through S, one of which is Q, contains p. The complete 8 by 8 truth table has exactly one simultaneous solution in each row and column. With the retained labels the matching is

$$(1,6),(26,37),(36,23),(61,34),(165,9),(167,17),(170,30),(174,41).$$

The first coordinate is a point label and the second a quadric index. Since the conditions use only incidence and G is transitive on conics, this proves (vi) for every conic. The actual actions on points and on quadrics are doubly transitive; the point stabilizer has quadric orbits of lengths 50 and 126. This also proves the source's transitivity assertions on incident and nonincident point–quadric pairs.

Coset maps, flags, and residues

For either family write H_1, H_2, H_3 in line, point, plane order. The retained base objects b_i have exactly these stabilizers in G. For every coset the certificate records the map

$$H_i g \longmapsto b_i^g. \tag{3}$$

It verifies each image literally and verifies the resulting six full bijections. Stabilizer equality makes (3) well-defined and equivariant. Two right cosets are incident when they intersect. The neighbours of H_i of type j are therefore the orbit of H_j under H_i. For all twelve ordered pairs of distinct types in the two families, the actual orbit is compared to the entire geometric neighbour set of b_i. Equivariance transports these equalities to all objects. This proves both incidence isomorphisms, without inferring incidence from matching degrees alone.

Every line has two points and two incident planes, and all four point–plane flags in its residue occur. Every incident point–plane pair extends through exactly seven lines. Thus every partial flag extends to a chamber, and both geometries have $15400 \cdot 4 = 61600$ chambers. Their base chamber stabilizers have order 720 and have explicit bijective generator maps onto Sym(6). Their G-orbits therefore contain all chambers. This is actual flag transitivity. The previously recognized residual connectivity, local primitivity, weak primitivity and intersection property transfer across (3).

For Γ_C, the base point residue has fifty conics as graph vertices and 175 lines as graph edges. All edges have distinct two-element endpoint sets. Breadth-first calculations at every vertex give valency seven, diameter two and girth five. The same calculation holds for the point and quadric residues of Γ_Q. The retained uniqueness statement identifies these graphs as Hoffman–Singleton graphs. The conic residue of Γ_C is literally its eight points and all their 28 pairs, hence K_8. Line residues are the complete bipartite graph on two points and two planes. Object transitivity transports the base-residue conclusions to all residues.

The faithful actions of the three order-252000 parabolics on their fifty-vertex residues have image order 252000. They therefore equal the full known automorphism group of the Hoffman–Singleton graph. The conic stabilizer's action equals Sym(8) and has trivial kernel. For each line stabilizer, the kernel on the four residue objects is exactly the already identified Borel S_6. Its image is elementary abelian of order four. The certificate supplies two commuting involutions generating an order-four subgroup disjoint from this kernel; together they generate the full line stabilizer. This proves the splitting asserted by the colon notation, in addition to the kernel and quotient orders.

The full automorphism and correlation groups

For Γ_C, restrict an automorphism to the bipartite point–conic incidence graph, with its two vertex classes distinguished. This restriction is faithful: a line is determined by its two points. Every graph

automorphism extends uniquely to lines, and the certificate checks all returned generators on every line–conic incidence. The complete graph calculation, on 1276 vertices, gives order 44 352 000. The explicit G-generators lie in that group and generate a faithful image of the same order, so equality holds. Since the three geometry types have distinct cardinalities, every correlation preserves types. Hence $\mathrm{Cor}(\Gamma_C) = \mathrm{Aut}(\Gamma_C) = G$.

For Γ_Q, use the point–quadric incidence graph, adjoining one marker for each type. The two markers form one separate colour class; each is joined to all 176 objects of its type. The 354-vertex graph therefore allows precisely type-preserving or type-interchanging permutations, not arbitrary permutations of incidence types. The complete group has order 88 704 000. Every returned generator is checked on all graph adjacencies. It has a unique lift to lines: if types are fixed, use the image point pair; if types are interchanged, use the inverse of the point-pair/quadric-pair bijection in (2). In both cases all 15400 resulting line incidences are checked, so these lifts are correlations of the full geometry. Conversely every correlation restricts to this graph, and the restriction is faithful.

Let D be the resulting group and let E be the explicit image of G. The exact generator map $G \to E$ is bijective. The kernel of D on the two markers equals E, its marker image has order two, and $C_D(E) = 1$. Thus conjugation embeds D in $\mathrm{Aut}(E)$; its order equals the retained full order of Aut(HS), so this embedding is surjective. Consequently the type-preserving group is G and the correlation group is Aut(HS). A saved generator interchanges the types and provides the duality claimed in [1, Section 4.2].

Certificate boundary

The finite files retain the complete object lists, both generator actions, six coset maps, twelve neighbour comparisons, the sixth-axiom matching, all small-graph generators and their line lifts, and explicit Borel and complement maps. Completeness of the graph-automorphism calculations is an acquired nauty premise; the returned generators are additionally checked literally. The GAP finite group and homomorphism algorithms and the retained identification of the ambient HS model are explicit dependencies. The original failed test of surjectivity onto the declared range of an action homomorphism is preserved. The successor uses its trivial kernel, the correct faithfulness assertion.

This proves the two construction and group descriptions in Section 4 of the fixed six-page article. It does not assert completion of the article's enumeration, Table 1, or historical supplement labels. Those obligations belong to their separately scoped components.

References

[1] D. Leemans, *The residually weakly primitive geometries of HS*, fixed six-page source, pp. 231–236, Section 4, pp. 233–235.

[2] G. Higman, *On the simple group of D. G. Higman and G. C. Sims*, Illinois J. Math. **13** (1969), 74–80. The historical attribution is retained as stated in [1].